Skewness and Pseudocodewords in Iterative Decoding

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I. Factor Graphs

Impressive performances by low-density parity-check codes and turbocodes have inspired researchers to determine how their iterative decoders depart from optimal decoding. Iterative decoding can be viewed as the application of the sum-product or min-sum algorithm in a graph that describes the constraints on a system of variables [1–3]. See [4] and [5] for contemporary monograph treatments.

Fig. 1a shows one possible *factor graph* [6] for a 3-bit repetition code C. Each check (filled disc) enforces a constraint (in this case, equality) on its neighboring codeword bits (unfilled discs). A configuration of the bits $\mathbf{x}$ is a valid codeword only if all checks are satisfied. (Note that this particular graph is not "minimal".)

II. Iterative Decoding Maximizes Correlation

Fig. 1b shows the *computation tree* [4] for 3 iterations of iterative decoding for bit 0. The most recent information passed to bit 0 comes from its three neighboring checks, which received information from bits 1 and 2, as shown in the top two levels of the tree. Earlier on during decoding, those bits received information from the bits shown at the next lower level, and so on.

It turns out that the iterative min-sum algorithm (the Viterbi form of turbodecoding) is equivalent to a noniterative min-sum algorithm on the computation tree. Let codewords $\tilde{\mathbf{x}}$ of the *tree code* $\tilde{C}$ be valid configurations of the bits in the *tree*. For each bit in the original code, there are many corresponding sites in $\tilde{C}$. Let $\ell_i(\tilde{\mathbf{x}})$ be the number of times that bit i in the original code has the value 1 in the tree codeword $\tilde{\mathbf{x}}$. For an AWGN channel with channel output column vector $\mathbf{y}$, the iterative min-sum algorithm computes

$$\tilde{\mathbf{x}}' = \operatorname*{argmax}_{\tilde{\mathbf{x}} \in \tilde{C}} \mathbf{y}^{\mathrm{T}} \boldsymbol{\ell}(\tilde{\mathbf{x}}), \tag{1}$$

where "T" indicates vector transpose. Corresponding to each tree codeword $\tilde{\mathbf{x}}$ is a vector in signal space $\boldsymbol{\ell}(\tilde{\mathbf{x}})$. The iterative min-sum algorithm finds the tree codeword that maximizes the correlation between this vector and the channel output. In contrast, the iterative sum-product decoder *averages* over the exponentiated negative $1/2\sigma^2$-weighted correlations between $\boldsymbol{\ell}(\tilde{\mathbf{x}})$ and the channel output.

III. Skewness in the Iterative Decoding Tree

Each bit i in the original code may have a different multiplicity m_i in the tree code, and as a result the correlation decoder is *skewed*. Let $f_i(\tilde{\mathbf{x}})$ be the *fraction* of times that bit i in the original code has the value 1 in the tree codeword $\tilde{\mathbf{x}}$: $f_i(\tilde{\mathbf{x}}) = \ell_i(\tilde{\mathbf{x}})/m_i$. From (1) the min-sum decoder computes

$$\tilde{\mathbf{x}}' = \operatorname*{argmax}_{\tilde{\mathbf{x}} \in \tilde{C}} \mathbf{y}^{\mathrm{T}} \mathbf{M} \mathbf{f}(\tilde{\mathbf{x}}), \tag{2}$$

where $\mathbf{M}$ is the diagonal matrix of multiplicities. By symmetry, the decoder should base its decision on $\mathbf{y}^{\mathrm{T}}\mathbf{f}(\tilde{\mathbf{x}})$. The multiplicity matrix skews the correlation decoder. By prescaling the channel outputs by the inverses of the multiplicities, we can correct for skewness.

For a given computation tree, the multiplicities can be computed using a recursion formula. For the 3-bit repetition code graph, each of x_1 and x_2 appears in the tree 0.9156 times as often as x_0 asymptotically. Fig. 1c shows the simulated BER for bit 0 as a function of a scale factor applied to the channel output for bit 0, for three different values of the AWGN E_b/N_0. The empirical optimal scale factor matches the theoretical prediction of 0.9156.

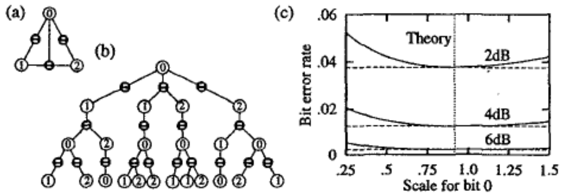

Fig. 1: (a) Factor graph for an $N = 3$ repetition code. (b) Tree code graph for 3 iterations terminating on bit 0 (the root). (c) Simulated BER for the 3-bit repetition code decoded by min-sum with the channel output for bit 0 scaled.

IV. Pseudocodewords in the Decoding Tree

In a repetition code, if x_i is 1 in the original codeword, all of its corresponding sites in the tree codeword will be 1. So, once corrected for skewness, the iterative decoder finds the *codeword* (not just tree codeword) that is maximally correlated with $\mathbf{y}$. When the original code and the tree code are not isomorphic, there will be tree codewords called *pseudocodewords* that have no analogue in the original code. There may be a pseudocodeword whose signal-space vector has higher correlation with the channel output vector than any of the true codewords. The skewness described in the previous section is caused by the graph alone, whereas the "skewness" due to pseudocodewords is caused by the local check functions.

After performing a fixed number of iterations, an iterative min-sum decoder decodes the root bit by comparing the lowest-cost tree codeword with a 0 in the root bit and the lowest-cost tree codeword with a 1 in the root bit. For a given channel output, these two codewords compete for the decision. The costs induced by the bits where these codewords differ will decide on the winner. Properties of the differences between these critical tree codewords cause iterative min-sum decoding to depart from ML decoding.

The factor graph given in [5] for a Hamming (7,4) code has pseudocodewords. If the channel outputs are corrected as described in Scn. III, the bit error rate becomes *worse*. This is because the critical tree codewords usually differ in bits that have low multiplicity in the tree, so that prescaling accentuates the wrong channel outputs.

V. Future Work

We are working on statistical and mathematical models of how pseudocodewords influence min-sum decoding. We hope these models will shed light on the behavior of sum-product decoding in factor graphs with cycles.

References

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